

IMAGINE Decoding Challenge: MEG Brain Signal Classification for Mental Imagery

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Abstract

We present the approach developed by team Immanuel_Varady_Bajaj_Saif for the IMAGINE Decoding Challenge on Kaggle, a 10-class brain-computer interface task hosted by Dr. Gordon Feld (Central Institute of Mental Health, Mannheim) as part of Johns Hopkins University's graduate Deep Neural Networks course. The competition requires predicting which of 10 imagined concepts (apple, bicycle, brush, cake, clown, cup, desk, foot, mountain, zebra) a participant is mentally visualizing from 306-channel MEG recordings, given only that classifiers must be trained on localizer (stimulus-evoked) epochs and applied to imagine epochs where subjects are performing mental imagery with eyes closed. The central challenge is a cross-brain-state domain gap: perception and imagery activate overlapping but distinct neural populations, making direct transfer difficult. Our key insight is that the localizer epoch contains a pre-visual auditory window (-1s to 0s) where subjects hear the target word before seeing the image - the same acoustic cue that initiates the imagine trial. Training on this auditory window and applying it to the imagine epoch's initial auditory response constitutes same-modality transfer, bypassing the perception-imagery domain gap entirely. We construct a 6-decoder weighted ensemble combining auditory and visual pathways, sliding-window OvR logistic regression banks, full-window classifiers, and GloVe-300 word embedding regression. Our final system achieves LB accuracy 0.133 (27th/169 teams, top 16%), up from a baseline SVC score of 0.108 (59th/92 teams) in week 2.

Key Results

1. Final LB accuracy 0.133, rank 27/169 (top 16%), 34 total submissions over 13 modules. Chance level 10%, prize threshold 20% (not reached by any team in our cohort).
2. Auditory-to-auditory same-modality transfer is the dominant signal. The auditory sliding decoder (weight 3.0) alone outperforms all cross-modal alternatives, confirming the theoretical prediction that same-acoustic-cue matching eliminates the perception-imagery gap.
3. Riemannian tangent-space covariance achieves 0.238 mean CV accuracy on localizer but collapses to 0.095 (chance) on imagery, demonstrating that spatial covariance structure does not transfer across brain states despite strong within-state performance.
4. Alpha-band log-variance (8-13 Hz, 1.0-3.0 s) performs at chance both cross-task and within imagery (5-fold CV 0.088), ruling out sustained alpha suppression as a usable class-discriminating feature at this

spatial and temporal scale.

5. GloVe word embedding regression contributes complementary signal: predicting a 300-dim semantic vector from MEG features and finding the nearest word by cosine distance encodes inter-class semantic geometry unavailable to flat 10-class logistic regression.

1. COMPETITION DESCRIPTION

The IMAGINE Decoding Challenge is a Kaggle competition designed around a fundamental question in cognitive neuroscience: can a classifier trained on brain responses to perceived stimuli (seeing an image, hearing a word) generalize to the brain state evoked by purely mental imagery of the same concept? The competition was hosted by Dr. Gordon Feld of the Central Institute of Mental Health, Mannheim, as a course project for Johns Hopkins University's graduate-level Deep Neural Networks (605.742) course, Spring 2026.

The task: predict, for each of 50 imagine trials per test subject, which of 10 word categories (apple, bicycle, brush, cake, clown, cup, desk, foot, mountain, zebra) the participant was mentally imagining. The prize pool of \$600/\$300/\$100 required accuracy $\geq 20\%$ (10% = chance). 169 teams participated.

1.1 Paradigm

Each subject participated in two phases inside an MEG scanner. In the localizer phase, they first heard a spoken word (auditory cue, approximately -1s to 0s relative to visual onset) and then saw an image of the corresponding object (visual window, 0 to +1s). This produced 480 labeled localizer epochs (48 per word class) per subject. In the imagine phase, subjects heard only the spoken word cue (0 to +1s) and then mentally imagined the object with eyes closed (imagery, +1s to +5s). Labels were scrambled and the task was to recover them. Crucially, the localizer auditory window and the imagine auditory window both capture responses to the same acoustic stimulus - this is the same-modality transfer pathway our system exploits.

1.2 Dataset

MEG was recorded with a MEGIN Triux (Vectorview layout): 306 channels total comprising 102 magnetometers and 204 planar gradiometers (plus ECG/EOG channels excluded from analysis). Data was downsampled to 100 Hz and stored in MNE Epochs format (1 epoch = 1 trial). Baseline correction was applied to localizer epochs using the -200ms to 0ms pre-onset interval.

Split	Type	N subjects	Epochs/subject	Time window	Labels
Train	Localizer	15	480 (48/class)	-200ms to +1000ms	Provided
Train	Imagine	15	50	-200ms to +5000ms	Provided (local CV)
Test	Localizer	14	480 (48/class)	-200ms to +1000ms	Provided
Test	Imagine	14	50	-200ms to +5000ms	Scrambled - predict

Table 1. Dataset summary. Only localizer data may be used for training; imagine test labels are the submission target.

2. PREPROCESSING

2.1 Channel Selection

We use only the 204 planar gradiometers (meg='grad' in MNE) after an ablation study showed that including the 102 magnetometers reduced validation accuracy. Magnetometers capture deep and distant field sources with lower spatial specificity, adding inter-subject noise without improving class discrimination at the single-trial level. The 2 ECG and 2 EOG channels are excluded as they do not reflect cortical activity.

2.2 Time Window Selection

Two analysis windows are defined per decoder pathway. The visual localizer window (0ms to 600ms post-visual onset) captures the category-discriminative visual evoked response; the first 200ms is a modality-nonspecific auditory onset response shared across all 10 categories (Hickok & Poeppel, 2007) and is therefore excluded. The auditory localizer window uses all available pre-visual time (approximately -1000ms to -50ms relative to visual onset). The imagine application windows mirror these boundaries: auditory 0ms-1000ms for the auditory pathway, 300ms-3000ms for the imagery pathway.

2.3 Null Data Injection

Pre-stimulus baseline epochs (before auditory cue onset) are randomly sampled and labeled as a null class during OvR classifier training. This sharpens the decision boundary between true neural responses and resting-state activity, reducing false positives when the sliding-window decoder scans the imagery epoch at time points that may not contain class-discriminating signal (Blankertz et al., 2011).

2.4 Rejected Preprocessing: ICA Artifact Removal

Independent Component Analysis (ICA) was initially evaluated to remove ECG and EOG artifacts via correlation-based component matching (Jung et al., 2000). Post-ICA models showed consistent performance degradation attributed to two factors: (1) signal loss - ICA components overlapping in space with artifact components can carry task-relevant neural activity (Castellanos & Makarov, 2006); (2) redundancy - the MEGIN system applies Signal Space Separation (SSS) preprocessing at acquisition time (Taulu & Simola, 2006), which already suppresses external interference. ICA was excluded from the final pipeline.

3. FEATURE ENGINEERING

Three feature families are computed from the gradiometer data and combined for the full-window and embedding regression decoders.

3.1 Temporal Window Features

The analysis epoch is divided into $N_{\text{windows}} = 5$ equal temporal segments. For each segment and each channel, the mean and standard deviation of the raw signal amplitude are computed, yielding $2 * 5 * 204 = 2,040$ features per epoch. This captures the coarse temporal dynamics of the evoked response while remaining robust to the trial-to-trial variability inherent in MEG data.

3.2 Band-Power Features

Frequency-band power is computed for four physiologically motivated bands via 4th-order Butterworth bandpass filtering followed by Hilbert envelope extraction. Mean power over the analysis window is averaged per channel, yielding $4 * 204 = 816$ additional features.

Band	Range (Hz)	Cognitive relevance
Theta	4 - 8	Working memory, attention, object recognition
Alpha	8 - 13	Inhibition, imagery suppression (8-13 Hz)
Beta	13 - 30	Motor planning, cognitive effort
Gamma	30 - 45	Local computation, perceptual binding

Table 2. Frequency bands used for band-power feature extraction.

3.3 GloVe Word Embeddings

Each of the 10 target word classes is represented by its 300-dimensional GloVe (Pennington et al., 2014) word vector, loaded from the glove-wiki-gigaword-300 model. These embeddings encode inter-word semantic geometry: semantically similar words (e.g., apple/cake) have higher cosine similarity than dissimilar pairs (e.g., mountain/zebra). The embedding regression decoder (Section 4.3) exploits this structure to place ambiguous predictions near semantically plausible classes rather than distributing errors uniformly.

3.4 Rejected Features: Riemannian Covariance

Riemannian tangent-space features (Ledoit-Wolf regularized covariance matrices projected to the tangent space of the SPD manifold via pyriemann, Barachant et al., 2012) achieved a mean 3-fold CV accuracy of 0.238 on the localizer alone, substantially above the 0.10 chance baseline. This confirmed the features capture genuine class-discriminating spatial covariance during visual perception. However, when applied to imagery, accuracy collapsed to 0.095 (range 0.020-0.160), indistinguishable from chance. The 14-point gap cannot be explained by the 21,000:480 feature-to-sample ratio alone - the decoder has capacity to overfit but instead generalizes at chance. The conclusion: spatial covariance structure during sustained imagery is qualitatively different from that during visual perception, and the Riemannian manifold learned on localizer data does not contain imagery-relevant directions. Riemannian features were excluded from the final ensemble.

4. MODEL ARCHITECTURE: 6-DECODER WEIGHTED ENSEMBLE

Our final system combines six independent decoders, each producing a ($N_{\text{trials}} \times 10$) probability matrix. The final prediction is a weighted average of these matrices followed by argmax. Decoder weights reflect the theoretical signal quality of each transfer pathway.

Decoder	Training Window	Application Window	Weight	Transfer Type
Auditory Sliding OvR	Loc. auditory (-1s to 0s)	Img. auditory (0s-1s)	3.0	Same-modality
Auditory Full-Window	Loc. auditory (-1s to 0s)	Img. auditory (0s-1s)	2.0	Same-modality
Auditory Embedding Reg.	Loc. auditory (-1s to 0s)	Img. auditory (0s-1s)	1.5	Same-modality
Visual Sliding OvR	Loc. visual (0s to 0.6s)	Img. imagery (0.3s-3s)	1.0	Cross-modal
Visual Full-Window	Loc. visual (0s to 0.6s)	Img. imagery (0.3s-3s)	0.5	Cross-modal
Visual Embedding Reg.	Loc. visual (0s to 0.6s)	Img. imagery (0.3s-3s)	0.5	Cross-modal

Table 3. Six-decoder ensemble configuration. Auditory decoders receive higher weights due to same-modality transfer.

4.1 Sliding Window OvR Bank

The core decoder trains a separate One-vs-Rest (OvR) L1-penalized logistic regression at every 10ms time step across the training window (50ms window width). At each step t , the 50ms $\times$ 204-channel segment is flattened to a $50 \times 204 = 10,200$ -dimensional vector. Per-class binary classifiers are trained: positive examples are trials of class c , negatives are all other trials plus null (pre-stimulus baseline) samples injected at a ratio of 48 null samples per class.

CV pruning: a 3-fold cross-validated accuracy check is run for each time window. Windows where CV accuracy falls below 11% (prune threshold) are discarded before inclusion in the bank. This removes time points that contain no class-discriminating signal, reducing noise in the ensemble aggregation. At inference, each kept window's logits are z-scored against a null distribution estimated from the training data's baseline samples, centered per trial, and aggregated by a weighted average proportional to each window's CV accuracy.

$$L_{\text{final}} = \sum_t \sum_d [w_d * cv_acc_t * \text{softmax}(z_score(\text{logits}_t, d))] \quad (Eq. 1)$$

4.2 Full-Window Classifier

A complementary decoder uses combined temporal + band-power features over the entire analysis window, reduced to 50 PCA components, followed by multinomial logistic regression ($C=1.0$, LBFGS solver, balanced class weights). At inference, the imagery epoch is scanned with 0.2s steps, and predicted probabilities are averaged across all valid window positions.

4.3 Word Embedding Regression

A Ridge regression ($\alpha=100$) is trained to predict the 300-dimensional GloVe vector of the target class from PCA-reduced MEG features. At test time, the predicted embedding is compared to the 10 target word vectors by cosine distance. A temperature-scaled softmax ($T=0.2$) converts cosine similarities to a probability distribution:

$$p_i = \exp(-d_i / T) / \sum_j \exp(-d_j / T), \quad d_i = \text{cosine_distance}(\text{pred}, \text{glove}_i) \quad (Eq. 2)$$

This decoder benefits from semantic geometry: even when the predicted embedding is inaccurate, it tends to

land near semantically related words rather than random ones. For instance, a confused 'apple' prediction may land closer to 'cake' than to 'zebra', partially recovering from the classification error.

5. RESULTS

5.1 Competition Leaderboard Performance

Module	LB Score	LB Rank / Teams	Cum. Submissions	Key Change
Week 2 (baseline)	0.108	59 / 92	2	Full MEG, SVC baseline
Week 3	0.114	46 / 103	13	Gradiometers only, window 200-1000ms
FPIP	0.120	36 / 118	24	Null injection, feature engineering
Week 10	0.133	16 / 142	27	Auditory pathway discovered
Week 13 (final)	0.133	27 / 169	34	Ensemble finalized

Table 4. Competition progress by module. Best rank 16th (0.133) at week 10; final rank 27th after more teams joined.

5.2 Ablation Study

We evaluated each decoder component in isolation on the 15 training subjects using within-subject cross-validation (localizer → imagine, label accuracy).

Decoder	Mean CV Accuracy	Median CV Acc.	Notes
Auditory Sliding OvR	0.141	0.140	Highest individual accuracy
Auditory Full-Window	0.128	0.120	Complementary temporal structure
Auditory Emb. Regression	0.119	0.100	Semantic geometry helps on confusables
Visual Sliding OvR	0.112	0.100	Cross-modal, harder transfer
Visual Full-Window	0.108	0.100	Near baseline individually
Visual Emb. Regression	0.106	0.100	Minor contribution; kept for diversity
Full 6-Decoder Ensemble	0.162	0.160	Best combined performance

Table 5. Per-decoder and ensemble accuracy on training subjects. Ensemble (0.162) substantially outperforms any individual component.

5.3 Failed Approaches

Two decoder architectures showed strong within-localizer performance but failed to transfer to imagery:

Riemannian Tangent-Space Decoder: Mean localizer CV accuracy 0.238 (range 0.146-0.450). Mean imagery accuracy 0.095 (range 0.020-0.160). The 14-point gap demonstrates that the learned covariance manifold does not generalize across brain states. Attempted remediation: (1) covariance recentering (subtracting task-specific mean before tangent projection); (2) occipital channel subset to reduce dimensionality. Neither approach recovered usable performance. Excluded from final ensemble.

Alpha-Band Decoder: Log-variance of the Hilbert envelope (8-13 Hz Butterworth, 1.0-3.0 s post-cue) per channel. Cross-task accuracy 0.084. To test whether the localizer-to-imagery transfer was the limiting factor, 5-fold CV was performed within imagery trials using the provided imagine-train labels. Within-imagery CV accuracy: 0.088 - indistinguishable from chance. The cross-task and within-imagery numbers being equal rules out the transfer hypothesis: the featurization simply contains no class-discriminating information at this band, time window, and spatial scale.

5.4 Subject-Level Failure Mode

Subject sub-26 exhibited a degenerate prediction: 48 of 50 imagine trials were classified as 'zebra'. This prediction collapse is a known failure mode in per-subject OvR classifiers when one class's training distribution overlaps strongly with the test distribution for a particular subject's brain geometry. Future mitigation: cross-subject regularization (constraining per-subject classifiers toward the population mean), or subject-specific recalibration using held-out imagine-train labels.

6. DISCUSSION AND FUTURE DIRECTIONS

The central lesson from this competition is that domain shift (the transfer gap between brain states) dominates over model complexity as the limiting factor in MEG decoding. A simple OvR logistic regression on the same-modality auditory pathway outperforms all cross-modal approaches regardless of their sophistication. This aligns with the neuroscience literature: Dijkstra et al. (2019) showed only partial overlap between visual perception and visual imagery in fMRI, and our MEG results suggest the overlap is even lower in the spatial covariance structure than in activation magnitude.

The GloVe embedding regression adds a qualitatively different type of signal: semantic structure. This is particularly valuable for confusable pairs - 'cup' and 'desk' are both indoor objects - where the embedding distance provides a soft prior that flat classifiers cannot exploit. The approach is related to the semantic embedding regression literature in fMRI (Mitchell et al., 2008) but applied here to MEG at a single-trial level.

Promising future directions include: (1) fine-tuning on labeled imagine-train trials to adapt the localizer-trained classifiers to the imagery distribution; (2) cross-subject transfer learning using subject-invariant features; (3) deep learning approaches (EEGNet, ConvLSTM) that may learn more transferable spatiotemporal representations than handcrafted features; (4) temporal attention mechanisms to automatically weight the most discriminative time points rather than relying on CV pruning.

7. REFERENCES

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